INCH-POUND

MIL-R-22097G
AMENDMENT 4
5 December 1997
SUPERSEDING
AMENDMENT 3
31 March 1996

MILITARY SPECIFICATION

RESISTOR, VARIABLE, NONWIRE WOUND
(ADJUSTMENT TYPE),
GENERAL SPECIFICATION FOR

This amendment forms a part of MIL-R-22097G, dated 26 November 1993, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1.1, line 5, delete "120°C" and substitute "125°C".

PAGE 2

TABLE I, note 2, delete and substitute:

"2. For characteristic C only, contact resistancevariation shall not exceed 3 percent or 20 ohms, whichever is greater. For characteristic F only, the contact resistance variation shall not exceed 3 percent or 3 ohms, whichever is greater."

PAGE 3 AND 4

* 2. APPLICABLE DOCUMENTS, delete and substitute:

"2. APPLICABLE DOCUMENTS

"2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. The section does not include documents cited in other sections of this specification or recommended for additional information or as samples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

"2.2 Government documents.

"2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (6.2).

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"SPECIFICATIONS

"DEPARTMENT OF DEFENSE

* "MIL-PRF-39035 - Resistor, Variable, Nonwire-Wound, (Adjustment Type), Established Reliability, General Specification For.

"STANDARDS

"DEPARTMENT OF DEFENSE

* "MIL-STD-202 - Test Methods Standard Electronics and Electrical Component Parts.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-1276 - Leads for Electronic Component Parts.

"(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Defense Automated Printing Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

"2.3 Non-Government publications The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

"AMERICAN NATIONAL STANDARD (ANS)

"J-STD-006 - Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications, Requirements for.

"(DoD activity may obtain copies of American National Standard J-STD-006 from the Standardization Document Order Desk. Private sector and other Government Agencies may purchase copies from the Institute for Interconnecting and Packaging Electronic Circuits (IPC), 7380 N. Lincoln, Lincolnwood, IL 60646).

"AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

"ANSI/NCSL Z540-1 - General Requirements for Calibration Laboratories and Measuring and Test Equipment.

"INTERNATIONAL ORGANIZATION FOR STANDARDS (ISO)

"ISO 10012-1 - Quality Assurance Requirements for Measuring Equipment . Part 1: Meteorological Confirmation System For Measuring Equipment.

"(Application for copies should be addressed to the American National Standards Institute, 11 West 42nd Street, New York, NY 10036).

"(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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"2.4 Order of precedence. In event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained."

PAGE 4

* 3.4, delete and substitute:

"3.4 Interface and physical dimension requirements. The resistors shall meet the interface and physical dimensions specified (see 3.1)."

PAGE 7

3.6, delete "$V = \sqrt{PR}$" and substitute "$E = \sqrt{PR}$ ".

PAGE 8

3.20, delete "Resistance between end terminals, with the contact arm in the position for setting", and substitute "Resistance between end terminals, with the contact arm in the position for setting stability".

PAGE 9

TABLE V, column 2, row 2, title, delete "%/°C" and substitute "ppm/°C".

PAGE 10

* 3.30.2.1, delete "Sn60 of QQ-S-571" and substitute "Type A of J-STD-006".

3.30.3, last line, delete "removing" and substitute "removed".

4.1.1, line 7, delete "eigher" and substitute "either".

4.1.2, last sentence, delete "MIL-C-45662" and substitute "ANSI/NCSL Z540-1, ISO 10012-1, or equivalent system, as approved by the qualifying activity.

PAGE 11

4.4.4b, delete in its entirety.

Paragraph following 4.4.4b, second sentence, delete "In addition to the periodic submission of inspection data,".

PAGE 15

4.5.2 through 4.5.2.1.1.1 inclusive, delete in their entirety.

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TABLE XI, delete in its entirety.

4.5.2.1.1.2 through 4.5.2.1.3 inclusive, delete in their entirety.

* 4.5.4, delete “MIL-R-39032” and substitute “section 5”.

4.6.4, delete in its entirety and substitute the following:

“4.6.4 Contact resistance variation (see 3.9) Contact resistance variation shall be measured with the measuring circuit shown on figure 3, or its equivalent. The operating shaft shall be rotated in both directions through 90 percent of the actual effective electrical travel for a maximum of six cycles. Only the last 3 cycles shall count in determining whether or not a contact resistance variation is observed at least twice in the same location, exclusive of the roll-on or roll-off points where the contact arm moves from the termination, on or off, the resistance element. Group A, subgroup I product acceptance may be determined based on one cycle minimum where compliance to the specification is demonstrated. The rate of rotation of the operating shaft shall be such that the wiper completes one cycle in 5 seconds minimum, to 2 minutes maximum.”

4.6.6, line 1, delete "301" and substitute "302".

4.6.7.2, at the end of the paragraph, add "(see 3.12.2)".

4.6.8c, at the end of the sentence, add "(see 3.13)".

4.6.10d(2), delete "For group B inspection: 10 cycles" and substitute "For group C inspection: 10 cycles".

4.6.16.2, delete in its entirety and substitute:

“4.6.16.2 Procedure. Total resistance shall be measured as specified in 4.6.2.1 and 4.6.8.1, respectively. The resistor shall be placed in a chamber at room temperature. The temperature shall be gradually decreased to -55°C +0°C, -5°C within a period off not less than one and one-half hours. For quality conformance inspection only, and at the option of the manufacturer, the resistors may be placed in the chamber when the chamber is already at the extreme low temperature. After 1 hour of stabilization at this temperature, setting stability shall be measured as specified 4.6.8.1. Full rated continuous working voltage (see 3.1 and 3.6) shall be applied for 45 minutes. The resistors may be loaded individually or in parallel. Fifteen +5, -0 minutes after removal of voltage, setting stability shall be measured as specified 4.6.8.1. The temperature in the chamber shall be gradually increased to room temperature within a period of not more than 8 hours. The resistors shall be removed from the chamber, and maintained at a temperature of 25°C ±5°C for period of approximately 24 hours. Total resistance shall be measured as specified 4.6.2.1. Resistors shall then be examined for evidence of mechanical damage.”

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PAGE 24

* 5. PACKAGING, delete and substitute:

"5. PACKAGING

"5.1 Packaging. For acquisition purposes, packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department, the Military Department's System Command, or the Defense Agency. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or responsible packaging activity."

PAGE 25

* 6.3, last two sentences, delete and substitute:

"The activity responsible for the QPL and for information pertaining to qualification of products may be obtained from Defense Supply Center, Columbus (DSCC-VQP), Post Office Box 3990, Columbus, OH 43216-5000".

PAGE 28

TABLE XV, column 4, title, delete "Tests from table IV to be performed on additional sample units" and substitute "Tests from table VI to be performed on additional sample units".

The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - AR, AT, AV, CR4, MI
Navy - AS, MC, OS
Air Force - 17, 19, 99

Preparing activity:
DLA - CC

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